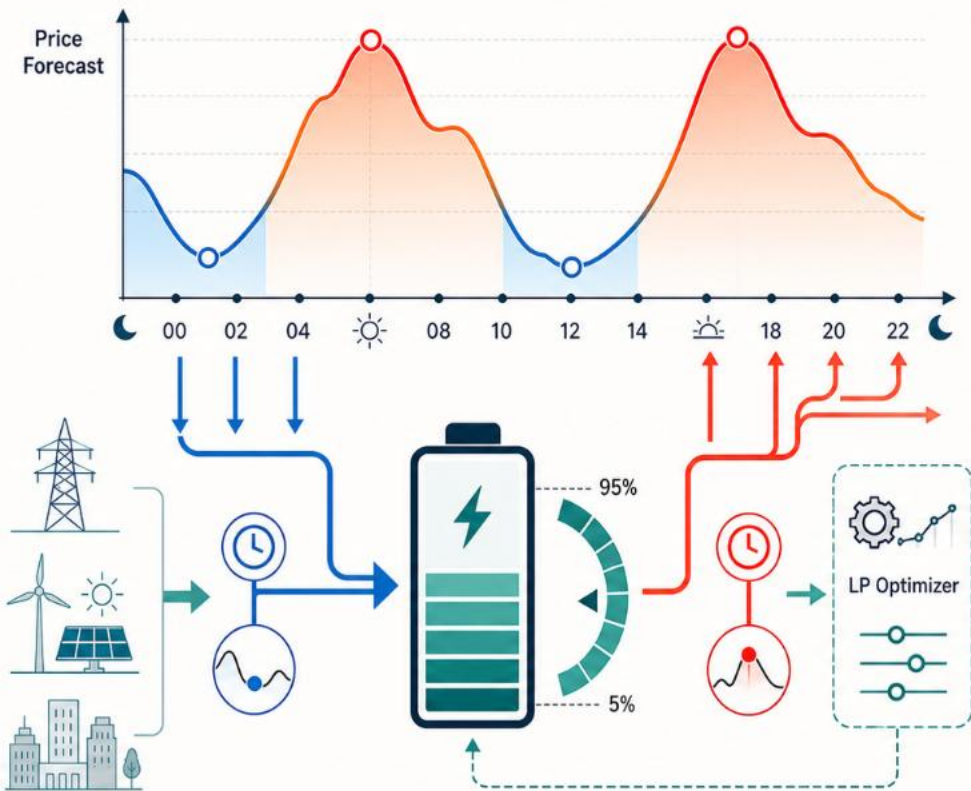




1. LP: Buy Low, Sell High

maximize
 $\text{price} * (\text{discharge} - \text{charge}) - \text{degradation}$



Charge when forecast price is low



Discharge when forecast price is high



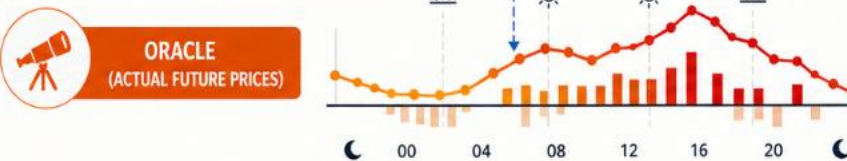
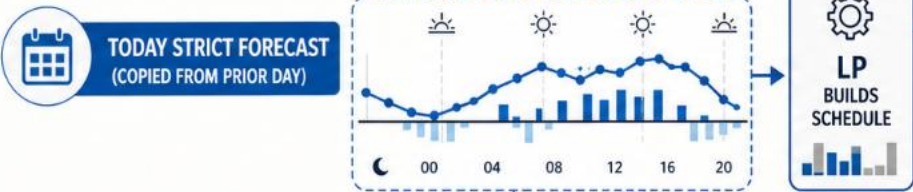
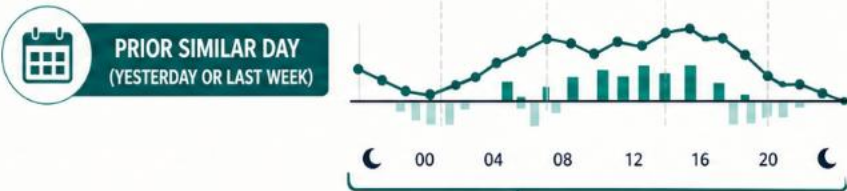
Respect SOC 5-95%, max power, efficiency



Only first hour is committed



Recommendation preview, not market execution



2. Strict Benchmark + Oracle

$$\text{regret} = \text{oracle value} - \text{achieved value}$$

 Strict forecast copies yesterday or last week

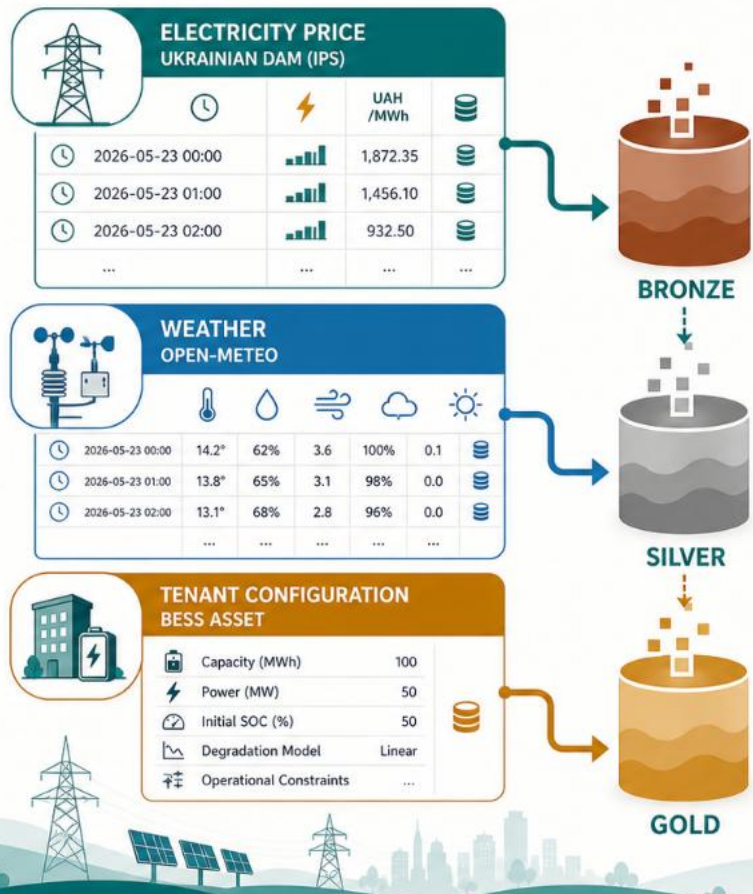
 LP builds a schedule from that forecast

 Oracle uses actual future prices

 Lower regret means better decision quality



Oracle is a scorer, not a live strategy



3. Raw Data

hourly row = timestamp + source + value + provenance



- OREE DAM: hourly IPS price, UAH/MWh



- Open-Meteo: hourly weather by tenant location



- Tenant config: capacity, power, SOC, degradation



- Synthetic fallback is demo-only

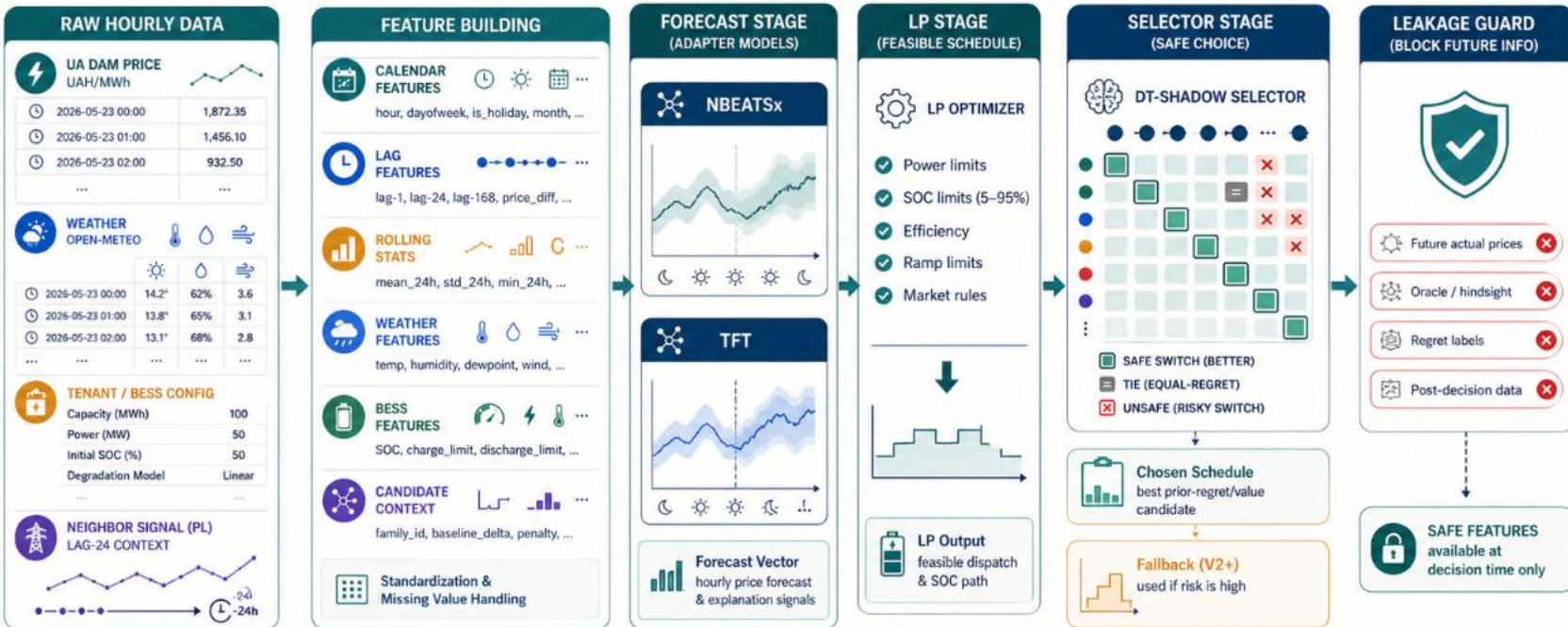


Same Ukrainian DAM price;
tenants differ by asset and weather

4. Features

features = calendar + lags + weather + BESS + candidate context

- ✓ Built only after source checks
- ✓ Forecast uses time, lags, rolling stats, weather
- ✓ LP uses forecast vector and battery limits
- ✓ Selector blocks future actual/oracle/regret leakage



Chosen by decision-time availability



Raw data by sources



Source checks & governance



Feature building



Forecast adapters



LP feasible schedule



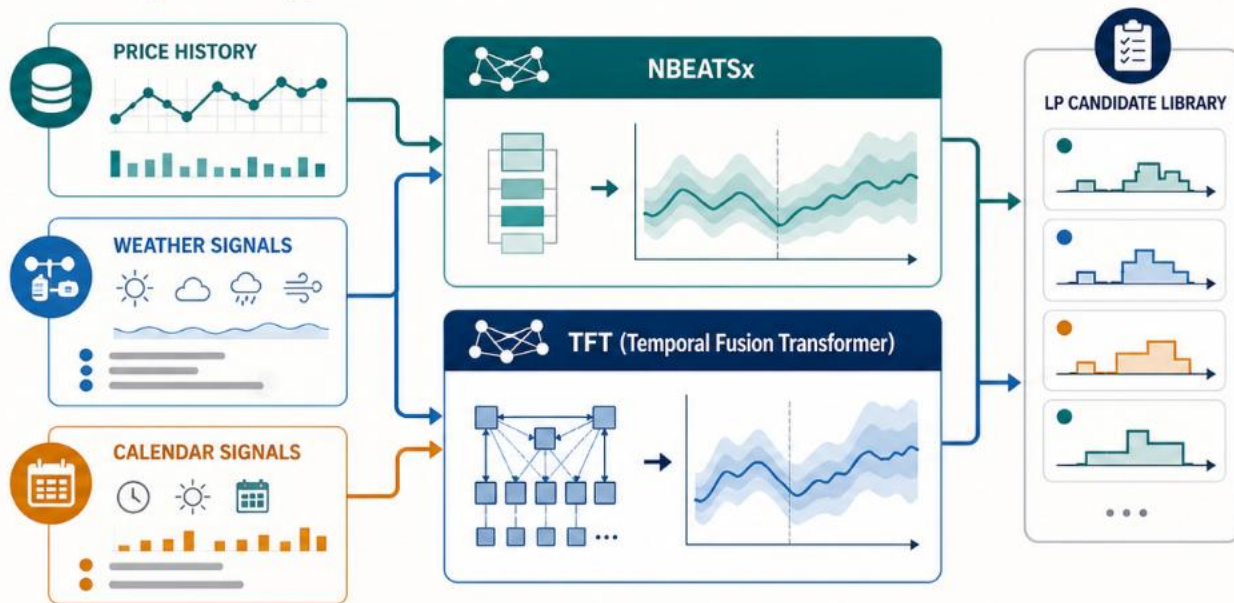
Selector safe choice



Leakage guard (decision-time)

5. TFT and NBEATSx

history + exogenous features → forecast → LP candidates



They are
forecast adapters



They do not
send market bids



TFT also gives
explanation signals



Scored by the same
LP/oracle evaluator

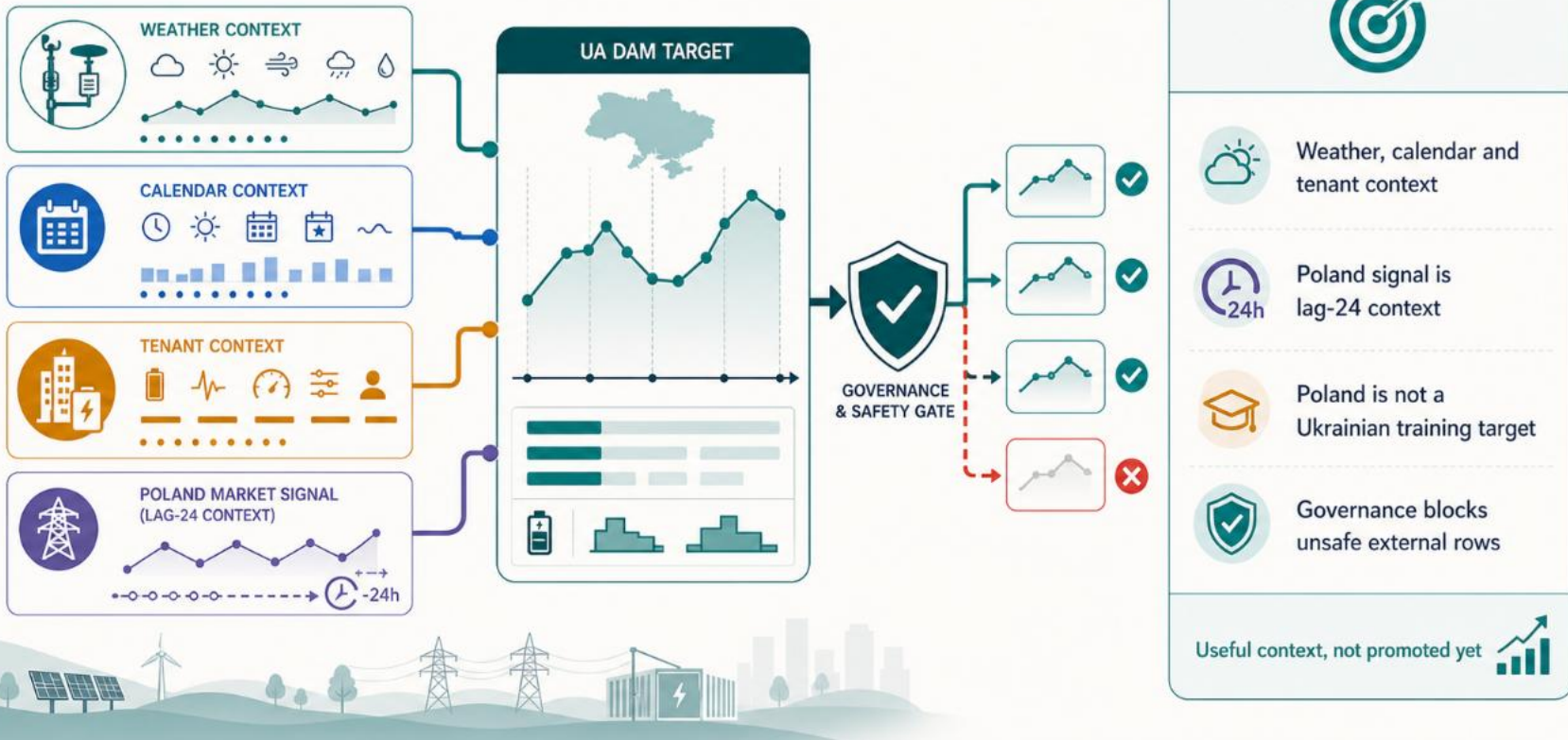


Forecast layer, not execution layer



6. Exogenous Features

UA DAM target + external context



7. Schedule Selection

candidate library → safety gate → selector → fallback



V2+ creates many feasible candidate schedules



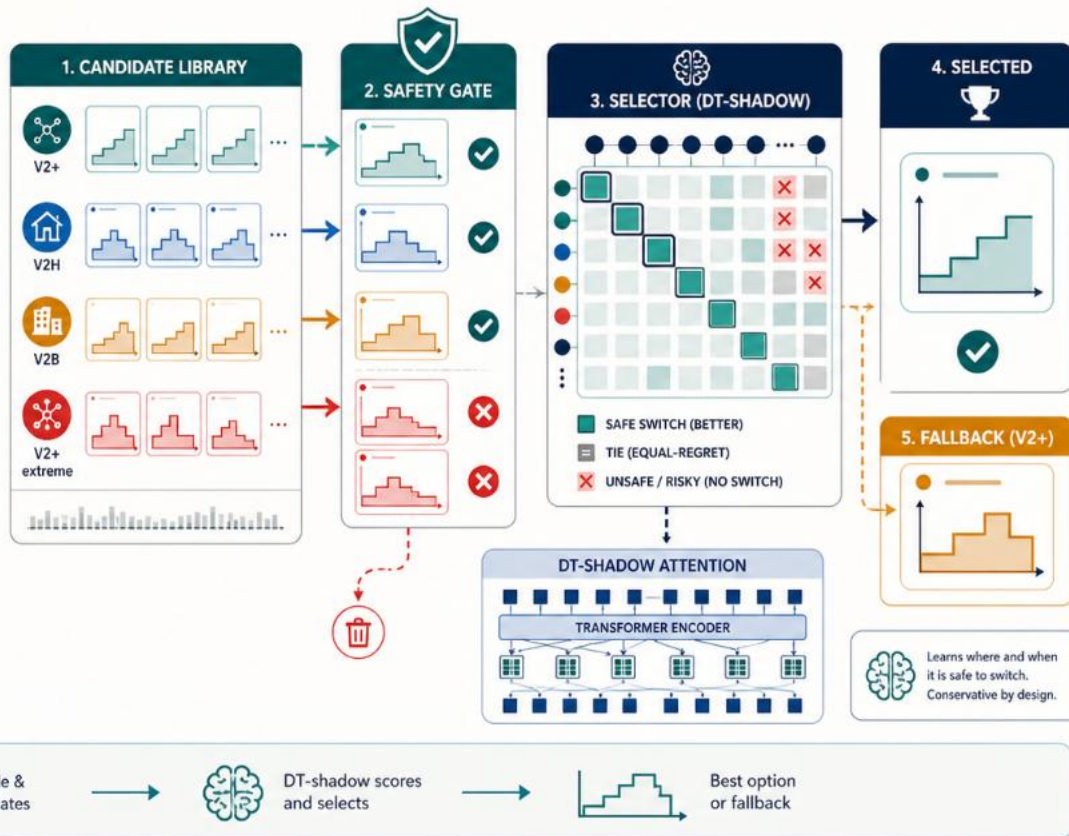
Families perturb or guard the baseline



DT-shadow predicts safe switches away from V2+



If risk is high, it abstains and keeps V2+



8. From Prediction to Recommendation

dispatch vector = hourly MW path + SOC path



Forecast vector goes into LP



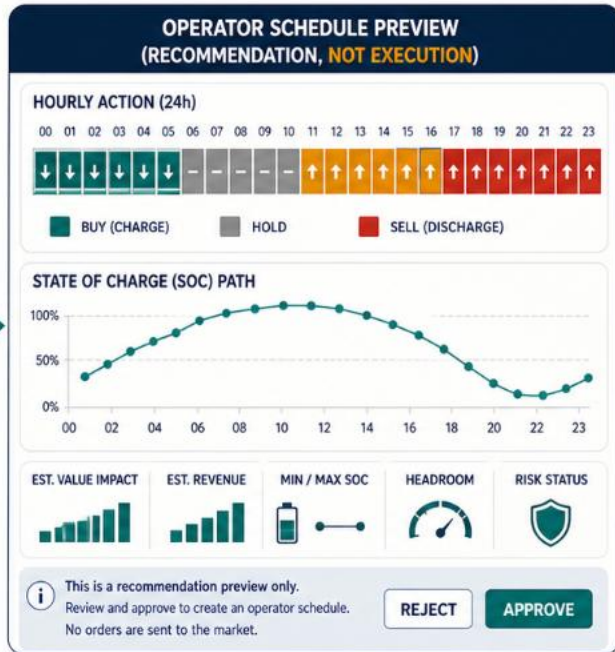
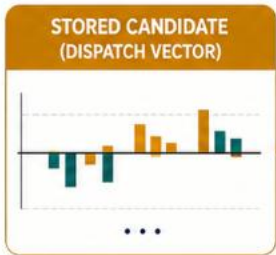
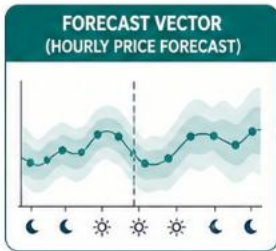
LP outputs feasible charge/discharge/SOC



Stored candidate reuses its dispatch vector



Operator sees BUY / HOLD / SELL preview



Recommendation schedule, not ProposedBid



09 9. Current DT Shadow Candidate



90 FINAL ANCHORS (EVALUATION BOARD)

	1	2	3	4	5	6	7	8	9	10	OUTCOME
1											
2						↻					↑ WIN
3											
4			↻								↑ WIN
5											
6								↻			↑ WIN
7											
8					↻						= TIE
9											

V2+ DEFAULT (KEPT)	SAFE SWITCH (MADE)	WIN (LOWER REGRET)	TIE (EQUAL REGRET)	LOSS (EQUAL REGRET)
86	4	3	1	0



mean regret:

174.77 → 168.16 UAH = 3.78% lower



90 final anchors: 86 stayed on V2+



4 safe switches to strict_reference



3 wins, 1 tie, 0 losses



Oracle scored the result afterward

REGRET COMPARISON (LOWER IS BETTER)



V2+ remains default; market_execution_enabled=false

